DSA BOOTCAMP 3.0 (C++ / JAVA)

Day 1: Basics of C++
(Data Types, Variables, Functions, Operators, Loops & Conditional statements/ Jump Statements, Syntax, Keywords etc)

Day 2 & 3: Arrays

- Introduction (Insert/ Access Elements)
- Reverse / Rotate Array
- Sum of Array
- Searching (Linear & Binary Search)
- Two Pointers
- Sorting (Quick / Selection / Insertion Sort)
- 2D Arrays

Day 4: Character arrays and strings

Day 5: Stacks & Queues (STL) + Basic OOPS (Theory Notes)

Day 6 & 7: Recursion

- Basic Ques (Fibonacci, factorial, sum of array)
- Generate Subsequence
- Coin toss
- Merge sort
- Quick sort

Day 8: Backtracking

- N Queens
- Rat In A Maze

Day 9 & 10: Linked List

- Insertion / Deletion From Head / Tail / Middle
- Find cycle, length of loop
- Middle element
- Kth Node from end
- Reverse a linked list

Day 12 & 13 : Trees

- Binary Trees Introduction
- Preorder + Postorder + Inorder Implementation
- Level Order Traversal
- Height / Diameter of Binary Tree
- Path Sum

Day 14: Binary Search Trees